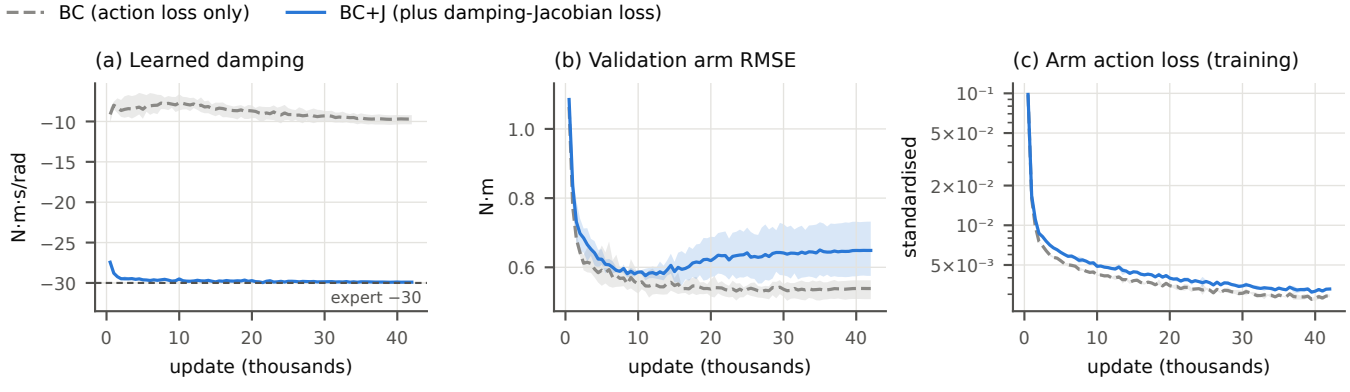
BC (action loss only)
BC+J (plus damping-Jacobian loss)
(a) Learned damping
N·m·s/rad
−10
−15
−20
−25
−30
expert −30
0
10
20
30
40
update (thousands)
(b) Validation arm RMSE
N·m
1.0
0.8
0.6
0
10
20
30
40
update (thousands)
(c) Arm action loss (training)
standardised
10⁻¹
5×10⁻²
2×10⁻²
10⁻²
5×10⁻³
0
10
20
30
40
update (thousands)